

War Risk Insurance / Shipping Capacity Shock Factor

Mathematical Framework for Capturing Equity Reratings from
Maritime Capacity Shocks and Insurance Withdrawal Events

Arka Labs – Quantitative Research

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Metric	Value
Overall Confidence (Thematic)	0.78
Overall Confidence (War Risk Standards)	0.88
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Table 1: Factor Summary Metrics

Signal Taxonomy Overview:

Signal	Description	Weight
A	Spot exposure (% fleet on spot / short charters)	0.45
B	War-risk pass-through (ability to reprice vs fixed charters)	0.25
C	Balance-sheet resilience (low leverage, liquidity)	0.20
D	Fleet quality/availability (modern fleet, utilization)	0.10
E	Avoid: charter-locked names with muted upside	-0.35
F1	Short leg: net jet-fuel exposure (airlines/travel)	-0.50
F2	Short leg: disruption sensitivity (airlines/travel)	-0.20

Table 2: Signal Taxonomy A–F with Composite Weights

Contents

1	Executive Summary	3
1.1	Factor Theme	3
1.2	Event Triggers	3
1.3	Geographic Scope	3
2	Universe and Eligibility	3
2.1	Long Universe Definition	3
2.2	Short Universe Definition (Optional)	4
2.3	Exclusions	4
2.4	Affected Segments	4
3	Signal Taxonomy and Scoring Methodology	4
3.1	Dimension A: Spot/Short-Charter Exposure	4
3.2	Dimension B: War-Risk Pass-Through Ability	4
3.3	Dimension C: Balance-Sheet Resilience	5
3.4	Dimension D: Fleet Quality and Availability	5
3.5	Dimension E: Charter-Lock Penalty (Avoid)	5
3.6	Dimension F1: Net Jet-Fuel Exposure (Airlines/Travel)	5
3.7	Dimension F2: Disruption Sensitivity (Airlines/Travel)	5
3.8	Composite Score Formula	6
4	Portfolio Weight Construction	6
4.1	Base Weight Formula	6
4.2	Weight Constraints	6
4.3	Constraint Application Order	6
4.4	Optimization Objective	7
5	Tradability Filters	7
6	Classification Tiers	7
7	War Risk Standards and Definitions	7
7.1	Issuing Authorities	7
7.2	War Risk Definition	8
7.3	Trigger Conditions	8
7.4	Coverage Types Affected	8
7.5	Key Notices and Effective Dates	8
8	Insurance Market Pricing and Withdrawal	8
8.1	Market Status	8
8.2	Observed War-Risk Rates	8
8.3	Key Circulars and Notices	9
8.4	Reinsurer Capacity	9
9	Market Rates and AIS Rerouting Metrics	9
9.1	Rate Index Reference	9
9.2	AIS Rerouting Methodology	9
10	Scenario Assumptions	10

11 Index Methodology and Tradability Thresholds	10
11.1 Index Providers Referenced	10
11.2 Inclusion Rule Examples	10
11.3 Liquidity Metrics	10
12 Academic and Empirical Evidence	10
12.1 Estimated Effect Size	10
12.2 Key Empirical Findings	11
12.3 Calibration Coefficients	11
13 Rebalancing Rules	11
13.1 Frequency	11
13.2 Trigger Conditions	11
13.3 Buffer Rules	11
13.4 Execution Procedure	11
13.5 Turnover Target	11
14 Constraints Summary	11
15 Validation Criteria	12
16 Execution Pipeline	12
16.1 Phase 1: Universe Selection and Profiles	12
16.2 Phase 2: Signal Extraction and Scoring	12
16.3 Phase 3: Aggregation and Portfolio Construction	13
17 Data Requirements	13
17.1 Fundamental Data	13
17.2 Market Data	14
17.3 Reference Data	14
18 External Sources	14
18.1 P&I Clubs and War Risk	14
18.2 Market Intelligence	14
19 Missing Data Requirements	15
20 Adapter Candidates	15
21 Implementability Notes	15
21.1 Phase 1: Pre-Identified Sources	15
21.2 Phase 2: Autonomous Research Confirmation	15
21.3 Phase 3: Gap Analysis	15
21.4 Phase 4: Adapter Mapping	16
21.5 Key Limitations	16
22 Step-by-Step Implementation Guide	16
23 Citations	16
24 Document Metadata	17
A Short Leg Candidates: Airlines/Travel Fields	17

B Fleet/Vessel Level Characteristics	18
C Industry/Company Exposure Fields	18

1 Executive Summary

1.1 Factor Theme

This thematic equity factor captures reratings from an **event-driven capacity shock**: withdrawal/cancellation of war-risk insurance, P&I/insurer exclusions and updated Listed Areas trigger vessels to reroute or pause transits (e.g., Red Sea, Bab el-Mandeb, Suez-related corridors), causing spot freight rates for crude, refined products, and LNG to spike as available tonnage tightens.

Long Universe: US-listed shipping (tankers, LNG carriers, marine services) with high spot/short-charter exposure should capture reratings.

Short Universe (Optional): US-listed airlines/travel companies with high jet-fuel sensitivity and route disruption exposure.

1.2 Event Triggers

The factor activates upon detection of the following events:

1. P&I or hull & machinery insurers issue formal Notice of Cancellation or withdrawal of war-risk cover for defined Listed Areas
2. Classification of a maritime route as “Listed Area” / “High-Risk” by P&I clubs or war-risk underwriters
3. Official maritime advisories (national maritime authorities / IMO / MARAD) warning of attacks or recommending avoidance
4. Surge in reported hostile incidents (missile/drone/ASBM attacks) against commercial vessels
5. Broker/market signals: rapid rise in spot war-risk premium or brokered war-risk surcharges
6. Sharp decline in AIS transit counts through a corridor (indicative of rerouting)

1.3 Geographic Scope

Primary geographic corridors affected:

- **Primary:** Red Sea / Southern Red Sea / Bab el-Mandeb / Gulf of Aden
- **Alternative Routing:** Suez Canal approaches and Cape of Good Hope
- **Contextual:** Black Sea / Turkish Straits (regional conflict)
- **Conditional:** Straits of Hormuz, Persian Gulf approaches (if escalation spreads)

2 Universe and Eligibility

2.1 Long Universe Definition

US-listed shipping equity names focused on crude/product tankers and LNG carriers plus marine services (shipowners, operators, ship management, broking houses) that:

1. Are US-listed (NYSE/NASDAQ)
2. Disclose fleet composition/charter mix in SEC filings and investor materials
3. Have meaningful spot or short-charter exposure (high % of fleet on spot or charters expiring <12 months)

2.2 Short Universe Definition (Optional)

US-listed airlines, major airfreight integrators, and travel & leisure companies with:

1. High sensitivity to jet-fuel as a percentage of operating costs
2. Material exposure to route disruption / increased fuel cost

2.3 Exclusions

- Charter-in/charter-out arbitrageurs fully charter-locked into long-term fixed charters with remaining durations $\gg$ shock horizon
- OTC or non-US-listed shipping names
- Names with opaque ownership/fleet data where chartering exposure cannot be inferred

2.4 Affected Segments

- Crude oil tankers (VLCCs, Suezmax, Aframax) transiting chokepoints
- Product/clean tankers carrying refined products on short-to-medium haul routes
- LNG carriers on flexible/spot cargoes (time-sensitive voyages)
- Marine services and voyage-dependent owners/operators
- (Short) Airlines, airfreight carriers, travel/tourism stocks

3 Signal Taxonomy and Scoring Methodology

3.1 Dimension A: Spot/Short-Charter Exposure

Definition: % of company-controlled fleet days exposed to spot or short-duration charters (e.g., charters expiring within 12 months).

Formula:

$$\text{spot_share}_i = \text{LLM_extract_pct}(\text{"\% fleet on spot or charters } \leq 6 \text{ months"}) \quad (1)$$

$$\text{score_A}_i = z_{\text{sector}}(\text{spot_share}_i) \quad (2)$$

Data Source: SEC filings (10-K/10-Q/6-K), investor presentations, company_overview profiles, voyage/fixture announcements, AIS-derived voyage patterns (proxy).

Weight in Composite: 0.45

3.2 Dimension B: War-Risk Pass-Through Ability

Definition: Operational ability to reprice voyages quickly—share of revenue mix from voyage/spot vs fixed time-charters; contractual pass-through clauses and ability to levy war-risk surcharges.

Formula:

$$\text{pass_through}_i \in [0, 1] \quad (1 = \text{strong ability to reprice}) \quad (3)$$

$$\text{score_B}_i = z_{\text{sector}}(\text{pass_through}_i) \quad (4)$$

Normalization: Map LLM labels {strong, moderate, weak, none} $\rightarrow$ {1.0, 0.5, 0.2, 0.0}, then z-score.

Weight in Composite: 0.25

3.3 Dimension C: Balance-Sheet Resilience

Definition: Liquidity and leverage: net leverage ratios, available liquidity, short-term maturities versus cash and revolver availability.

Formula:

$$\text{bs_resilience}_i = 0.7 \times \text{LLM_score_low_leverage}_i + 0.3 \times \text{LLM_score_liquidity}_i - 0.2 \times z(\text{days_to_cover}_i) \quad (5)$$

where $\text{days_to_cover}_i = \text{short_interest}_i / \text{avg_daily_volume}_i$

Weight in Composite: 0.20

3.4 Dimension D: Fleet Quality and Availability

Definition: Average fleet age, % modern (eco-design) vessels, owned vs chartered capacity, and historical utilization/idle days.

Formula:

$$\text{fleet_quality}_i = 0.6 \times (-z(\text{fleet_avg_age}_i)) + 0.2 \times z(\text{scrubber_pct}_i) + 0.2 \times z(\text{availability}_{\leq 90d,i}) \quad (6)$$

$$\text{score_D}_i = z_{\text{sector}}(\text{fleet_quality}_i) \quad (7)$$

Weight in Composite: 0.10

3.5 Dimension E: Charter-Lock Penalty (Avoid)

Definition: Flags for names with high proportion of fleet on long-term fixed charters or bare-boat contracts that prevent near-term spot upside.

Formula:

$$\text{penalty_E}_i = 0.35 \times z_{\text{sector}}(\text{long_TC_share}_{>12m,i}) \quad (8)$$

$$\text{contribution} = -\text{penalty_E}_i \quad (9)$$

Weight in Composite: -0.35

3.6 Dimension F1: Net Jet-Fuel Exposure (Airlines/Travel)

Definition: For shorts: airlines/travel sensitivity measured by fuel cost as % of operating costs and forward fuel exposure.

Formula:

$$\text{net_fuel_exposure}_j = \text{fuel_cost_share}_{\%j} \times (1 - \text{hedge_ratio}_{\%_next12m_j}) \quad (10)$$

$$\text{contribution} = -0.50 \times z_{\text{sector}}(\text{net_fuel_exposure}_j) \quad (11)$$

Weight in Composite: -0.50

3.7 Dimension F2: Disruption Sensitivity (Airlines/Travel)

Definition: Long-haul share, Middle East/Europe-Asia reliance, and operational fragility indicators.

Formula:

$$\text{disrupt_sens}_j \in [0, 1] \quad \text{from LLM classification} \quad (12)$$

$$\text{contribution} = -0.20 \times z_{\text{sector}}(\text{disrupt_sens}_j) \quad (13)$$

Weight in Composite: -0.20

3.8 Composite Score Formula

For any stock i :

$$\begin{aligned} \text{Score}_i = & \mathbf{1}_{\text{shipping}} \left(0.45 \times \text{score_A}_i + 0.25 \times \text{score_B}_i + 0.20 \times \text{bs_resilience}_i \right. \\ & \left. + 0.10 \times \text{score_D}_i - 0.35 \times z(\text{long_TC_share}_{>12m,i}) \right) \\ & + \mathbf{1}_{\text{airlines/travel}} \left(-0.50 \times z(\text{net_fuel_exposure}_i) - 0.20 \times z(\text{disrupt_sens}_i) \right) \end{aligned} \quad (14)$$

Cohorts are standardized separately each rebalance; final Score_i is re-standardized across the combined eligible universe.

Score Range: Approximately $[-3.5, +3.5]$ after winsorization; typical middle 80% within $[-2, +2]$.

4 Portfolio Weight Construction

4.1 Base Weight Formula

Let $S_i = \text{Score}_i$ after cross-sectional z-score and winsorization.

Define positive leg $L = \{i : S_i > 0\}$ and negative leg $K = \{i : S_i < 0\}$.

Pre-liquidity weights:

$$w_i^+ = \frac{S_i}{\sum_{j \in L} S_j} \quad \text{for } i \in L \quad (15)$$

$$w_i^- = \frac{|S_i|}{\sum_{j \in K} |S_j|} \quad \text{for } i \in K \quad (16)$$

Liquidity adjustment:

$$\text{liq}_i = \min \left(1, \frac{\text{ADV}_i}{10,000,000} \right) \quad (17)$$

Adjusted weights:

$$w_i^{+*} = w_i^+ \times \text{liq}_i \quad ; \quad w_i^{-*} = w_i^- \times \text{liq}_i \quad (18)$$

Target gross = 1.0 with long = short = 0.5 (beta-neutral scaling may rebalance leg totals).

4.2 Weight Constraints

Constraint	Value
Minimum Weight	0.5%
Maximum Weight (Single Name)	7.0%
Shipping/Marine Long Leg Max	95% of gross
Airlines/Travel Short Leg Max	95% of gross
Gross Exposure Target	1.0
Net Exposure Target	0.0 ± 0.02
Portfolio Beta Target	$[-0.05, +0.05]$

Table 3: Portfolio Weight Constraints

4.3 Constraint Application Order

1. Apply liquidity adjustment
2. Clip to $[\text{min}, \text{max}]$ bounds

3. Enforce single-name cap (7%)
4. Renormalize within legs
5. Beta-neutralize vs SPY (solve for scalar α on short leg such that $\beta_{\text{long}} - \alpha \cdot \beta_{\text{short}} = 0$)
6. Final renormalize to gross = 1.0

4.4 Optimization Objective

Closed-form direct characteristic portfolio with post-constraint repair. Optional mean-variance overlay to minimize ex-ante variance of active returns subject to keeping target exposure to S_i , using realized covariance from trailing 252 trading days.

5 Tradability Filters

Filter	Threshold
Minimum ADV (63-day)	\$5,000,000 USD
Minimum Market Cap	\$500,000,000 USD
Maximum Bid-Ask Spread	60 bps
Minimum Price	\$5.00
Excluded Exchanges	OTC, Pink Sheets
Excluded Security Types	ADR, ETF, Preferred, CEF
Participation Cap	$\leq 10\%$ of 20-day ADV per name

Table 4: Tradability Filter Thresholds

6 Classification Tiers

Tier	Score Range	Weight Range	Characteristics
Pure Play Shipping Beneficiary	[1.5, 3.5]	[3%, 7%]	High spot share, strong pass-through, low charter lock, healthy BS, younger/available fleet
Significant Exposure	[0.5, 1.5]	[1%, 3%]	Meaningful spot leverage or pass-through with acceptable BS/availability
Neutral or Mixed	[-0.5, 0.5]	[0%, 1%]	Balanced features; low conviction
Short Candidate (Fuel/Disruption)	[-3.5, -0.5]	[-3%, -7%]	High jet-fuel exposure net of hedges and/or high disruption sensitivity

Table 5: Classification Tiers by Score Range

7 War Risk Standards and Definitions

7.1 Issuing Authorities

- P&I Clubs: Skuld, UK P&I, Gard, West of England, Japan P&I, American Club
- International Group of P&I Clubs / Joint War Committee (LMA/JWLA)
- BIMCO (voyage/time charter clauses)
- IMO, national maritime authorities (e.g., US MARAD)

7.2 War Risk Definition

From P&I Rule 58: “The Association shall not cover under a P&I entry liabilities, losses, costs or expenses . . . when the loss . . . was caused by: war, civil war, revolution, rebellion, insurrection or civil strife.”

7.3 Trigger Conditions

1. Presence/operation within a published “Listed Area” (club/JWLA coordinates)
2. Hostilities or hostile acts by/against belligerent powers
3. Capture, seizure, arrest, restraint or detainment
4. Derelict mines/torpedoes/bombs or other weapons of war
5. Terrorism or politically-motivated malicious acts
6. Blockades, confiscation, nationalization, requisition
7. Reinsurer withdrawal of support for specified zones

7.4 Coverage Types Affected

- P&I War Risk cover (non-mutual/non-poolable war extensions)
- Charterers’ Liability cover and Non-pooled extended ancillary covers
- Hull & Machinery and Loss of Hire policies (separate war policies apply)
- Excess War Risks Insurance (EWRI) / buy-back policies
- Mutual P&I entries / Group Excess P&I war covers

7.5 Key Notices and Effective Dates

Example from authoritative circulars: **Noon GMT 20 February 2024** (Skuld / UK P&I Notices of Cancellation for Red Sea / Bab el-Mandeb / Gulf of Aden).

8 Insurance Market Pricing and Withdrawal

8.1 Market Status

Selective market appetite with active underwriting withdrawals for non-mutual war-risk lines; buy-back (charterers’ liability / P&I excess) available from some clubs; market showing highly elevated short-term Additional Premium (AP) pricing for Red Sea/Bab el-Mandeb/Gulf of Aden transits.

8.2 Observed War-Risk Rates

Zone	Rate (% Hull)	Validity	Notes
Red Sea / Bab el-Mandeb / Gulf of Aden	0.7% – 1.0%	7 days	Common market construct; upper band during late-2025/early-2026 spikes
Red Sea (low-volatility baseline)	0.2%	7 days	December 2025 ceasefire/low incidence period

Table 6: War-Risk Additional Premium (AP) Surcharge Table

Uplift from Baseline: ~400% (from 0.2% to 0.7–1.0%)

Rate Basis: Additional Premium (AP) charged as percent of hull value for short (typically 7-day) coverage.

Observation Date: 2026-01-28T08:00:00 UTC

8.3 Key Circulars and Notices

- Circular No. 33/22 – Notice of Cancellation in Respect of War Risks (American Club)
- 21/2023: Red Sea Buyback – Charterers’ Liability Cover (Gard)
- Circular 03/24: War Risks P&I Excess Cover (UK P&I)
- Buy-back cover available in respect of certain war risks (Skuld)

8.4 Reinsurer Capacity

Reinsurers have issued notices of cancellation for certain war-risk lines, causing a material capacity squeeze and forcing clubs/insurers to either cancel non-mutual cover or offer buy-back cover at materially higher prices; market capacity varies by flag and vessel linkage (US/UK-linked vessels face selective refusal by some underwriters).

9 Market Rates and AIS Rerouting Metrics

9.1 Rate Index Reference

Index Name: Baltic Exchange headline & route assessments (VLCC TD3C, BDTI/BCTI) + broker spot assessments (Clarksons / Braemar)

Vessel Types: VLCC TD3C (AG-JPN), Suezmax, Aframax, Product Tanker (MR/Handy), LNG (170k/174k CBM)

Quote Unit: USD/day (for time-charter/spot), WS points (voyage charter), USD/ton

9.2 AIS Rerouting Methodology

Two-part construct:

1. **Market-rate time series:** Collect broker/Baltic headline route assessments and normalize to USD/day per vessel-type and route
2. **AIS rerouting/capacity metrics:** Use commercial AIS history (6–12 month backfill) to detect:
 - Route deviations (planned vs actual track)
 - AIS “dark” gaps (periods where AIS disabled near threat corridors)
 - Additional distance/time on rerouted voyages

Rerouted Percent Formula:

$$\text{rerouted_pct} = \frac{\# \text{ vessels that deviated from standard corridor}}{\text{baseline } \# \text{ vessels over equivalent prior window}} \quad (19)$$

Reroute Threshold: added_nm > 500 nautical miles or AIS track leaves Suez corridor polygon

10 Scenario Assumptions

Parameter	Baseline	Severe
Duration (days)	30	90
Listed Area Scope	Red Sea + Bab el-Mandeb + Gulf of Aden	+ Suez approaches + expanded
Percent Rerouted	40%	70%
Additional Days per Voyage	10	20
Premium Uplift (%)	100%	200%

Table 7: Scenario Assumptions for Capacity Shock Modeling

Parameters are configurable; percent_rerouted and additional_days should be validated against AIS-derived transit counts and broker freight indices.

11 Index Methodology and Tradability Thresholds

11.1 Index Providers Referenced

- MSCI GIMI / FTSE GEIS (Global All Cap / Micro)
- S&P Dow Jones Indices (Composite 1500 / Total Market)
- Nasdaq-100 (NDX)
- Baltic Exchange (BDTI, BDI, BLNG)

11.2 Inclusion Rule Examples

MSCI: “A minimum liquidity level of 20% of 3-month ATVR and 90% of 3-month Frequency of Trading over the last 4 consecutive quarters, as well as 20% of 12-month ATVR are required.”

FTSE Russell (GEIS): “A non-constituent must pass at least 10 out of 12 months with a monthly median turnover of at least 0.050% of their shares in issue.”

S&P DJI: “FALR must be ≥ 0.1 (Total Market); FALR must be ≥ 0.75 (Composite 1500 addition).”

Nasdaq NDX: “A security must have a three-month average daily traded value of at least \$5 million (USD).”

11.3 Liquidity Metrics

Metric	Threshold	Source
FTSE Median Monthly Turnover	0.05%	FTSE GEIS
ADV Minimum (USD)	\$5,000,000	Nasdaq/Internal
Market Cap Floor (USD)	\$2,000,000,000	Index Standard

Table 8: Liquidity Threshold Reference

12 Academic and Empirical Evidence

12.1 Estimated Effect Size

Median Short-Term Spike: 25% change in route-level spot/TCE or freight rate

Confidence Interval: [8%, 60%]

Outcome Metric: Route- and vessel-class spot freight rate (TCE or spot charter rate) measured as percent change relative to pre-event baseline (30-day prior average).

12.2 Key Empirical Findings

- Route-level cost estimates: transport-chain cost increases $\sim 3\%$ – 13% (Journal of Transport Geography)
- Voyage-level rerouting costs: millions USD per voyage for VLCC/container vessels (Bendall)
- Geopolitical risk has persistent impacts on freight indices (Energy 2023)
- LNG vessels: ~ 21 day round-trip additions and inter-basin premium documented

12.3 Calibration Coefficients

Coefficient	Value	Notes
α_A (Spot Exposure Weight)	0.45	Highest weight; strong pass-through to freight
α_B (War-Risk Pass-Through)	0.25	Ability to reprice on spot vs fixed
α_C (Balance-Sheet Resilience)	0.15	Penalty for high leverage / low liquidity
α_D (Fleet Quality/Availability)	0.15	Modern fleet / younger tonnage increases optionality

Table 9: Calibration Coefficients from Empirical Evidence

13 Rebalancing Rules

13.1 Frequency

Monthly on the first US trading day.

13.2 Trigger Conditions

- **Event Trigger:** War-risk market regime change (operator flag) — run ad-hoc rebalance within $T+1$
- **Urgent Risk:** Name-level ADV drops below \$3M or price $< \$5$ — remove at next trading close

13.3 Buffer Rules

5% entry/exit score buffer around long/short cutoffs; maintain names unless Score_i crosses 0 by more than 0.25σ to reduce churn.

13.4 Execution Procedure

Trade at VWAP over 2–3 days post-rebalance; stagger low-ADV names; enforce participation cap $\leq 10\%$ of 20-day ADV per name.

13.5 Turnover Target

Annual turnover $\leq 200\%$ gross; alert if exceeded two consecutive months.

14 Constraints Summary

- Gross exposure = 1.0; target net exposure = 0.0 ± 0.02 (beta neutrality: portfolio $\beta \in [-0.05, +0.05]$)

- Single-name weight cap 7%; minimum live weight 0.5% (else drop)
- Liquidity: participation $\leq 10\%$ of 20-day ADV; no new positions where ADV < \$5M
- Cohort bounds: shipping/marine-services long leg $\leq 95\%$ of gross; airlines/travel short leg $\leq 95\%$ of gross
- Turnover control: 20% name-level turnover cap per month unless event-triggered
- No OTC/Pink listings; exclude ADRs/ETFs/Prefs/CEFs

15 Validation Criteria

1. **Backtest window:** At least 2010-01-01 to most recent month-end; form monthly portfolios
2. **Benchmark comparisons:** (a) SPY; (b) equal-weighted shipping vs airlines baskets
3. **Risk/return metrics:** Annualized return, volatility, Sharpe, Sortino, max drawdown, hit rate
4. **Exposure hygiene:** Regression on Fama–French 5 + momentum; report loadings and t-stats
5. **Liquidity & capacity:** Average and 95th percentile participation, median ADV coverage
6. **Stability tests:** Sensitivity to ± 2 changes in weight caps; varying cohort z-score windows
7. **Alpha combination robustness:** Test equal-weight vs characteristic-weighted scores
8. **Construction robustness:** Verify targeted exposures retained pre/post constraints

16 Execution Pipeline

The factor execution consists of multiple phases across thematic context, war risk standards, insurance pricing, market rates, and factor methodology. Below is the consolidated execution pipeline.

16.1 Phase 1: Universe Selection and Profiles

Step	Agent	Adapter	Description
1	UniverseSelectionAgent		Load S&P 500 + Russell 2000; filter for shipping/marine services and airlines/travel
2	GenericExtractionAgent	company_overview	Fetch company profiles (industry, sector, description, CIK)
3	GenericExtractionAgent	sec_adapter	Pull SEC filings (10-K/10-Q/6-K/8-K) for past 3 years
4	ClassificationAgent	—	Classify filings for charter types, war-risk clauses, fleet age, debt/cash

16.2 Phase 2: Signal Extraction and Scoring

Step	Agent	Adapter	Description
5	LLMScoringAgent	—	Derive spot_exposure_pct, charter_locked_flag, war_risk_pass_through_score

Step	Agent	Adapter	Description
6	GenericScoringAgent	sec_adapter	Extract balance sheet metrics (net leverage, current ratio, liquidity buffer)
7	GenericExtractionAgent	sec_adapter	Extract fleet tables (vessel type, year built, owned vs chartered)
8	GenericExtractionAgent	polygon_adapter	Fetch daily close prices, ADV, short interest
9	GenericExtractionAgent		Monitor external triggers (P&I circulars, MARAD, broker reports, AIS)

16.3 Phase 3: Aggregation and Portfolio Construction

Step	Agent	Adapter	Description
10	GenericAggregationAgent		Combine signals into composite score with configurable weights
11	GenericScoringAgent	—	Apply cohort-wise z-normalization and composite score formula
12	CodeGeneratingAgent	—	Compute OLS betas vs SPY, shrinkage covariance
13	GenericScoringAgent	—	Compute base weights from scores; apply liquidity adjustment and clipping
14	GenericAggregationAgent		Beta-neutralize and finalize weights (gross = 1.0)
15	OutputAgent	—	Generate rebalance pack, audit trail, factor loadings

17 Data Requirements

17.1 Fundamental Data

Source	Fields	Frequency	Used By
sec_adapter.get_filings	Full-text 10-K/20-F/6-K/10-Q/8-K: fleet tables, charter mix, TCE, leverage, fuel/hedging	Quarterly	A,B,C,D,E,F1,F2
fmp_adapter.get_earnings.transcripts	Prepared remarks + Q&A; spot/TCE, pass-through, redelivery commentary	Quarterly	A,B,C,D,F2

Table 13: Fundamental Data Requirements

17.2 Market Data

Source	Fields	Frequency	Used By
polygon_adapter.avg_dollar_volume	Average dollar volume (63 trading days)	Each rebalance	Tradability, liquidity adj
polygon_adapter.get_daily_close_prices	Daily closes for tickers and SPY	Daily (252d)	Beta, risk model
polygon_adapter.get_short_interest	Settlement date, short interest, days to cover	Biweekly	C (resilience)

Table 14: Market Data Requirements

17.3 Reference Data

Source	Fields	Frequency	Used By
company_overview.get_company_profile	company_name, industry, sector, description	Quarterly	Universe filters
company_overview.check_topic_relevance	Binary/LLM answer for topics	On demand	Universe preselection

Table 15: Reference Data Requirements

18 External Sources

18.1 P&I Clubs and War Risk

Domain	Data Types	Notes
skuld.com	P&I circulars, Designated Area maps	Notice of Cancellation texts, Listed Area coordinates
gard.no	Club circulars, Rule 58 guidance	Canonical P&I exclusion text
ukpandi.com	Circulars, board resolutions	P&I guidance on supplementary cover
bimco.org	VOYWAR 2025, CON-WARTIME 2025	Contractual war-risk clause templates
maritime.dot.gov	MARAD advisories	US national guidance on threats
imo.org	IMO circulars/advice	Operational routing guidance

Table 16: P&I and Regulatory External Sources

18.2 Market Intelligence

Domain	Data Types	Notes
balticexchange.com	Freight indices (BDTI, BDI, BLNG)	Route assessments, index definitions
shippingintelligencehub.com	War-risk premium reporting	Market bulletins, observed AP quotes
marinetraffic.com / spire.com	AIS positions, vessel status	Rerouting and availability inference
clarksons.com	Fleet databases, TCE estimates	Proprietary/subscription
lloydslistintelligence.com	AIS gaps, market analysis	Paywalled intelligence

Table 17: Market Intelligence External Sources

19 Missing Data Requirements

1. **Vessel-level charter status:** Comprehensive company-level %fleet-on-spot / short-charter not typically filed in machine-readable format
2. **War-risk premium levels:** Granular by route and vessel class (usually from brokers/insurers; not in adapters)
3. **AIS/vessel routing data:** High-resolution AIS and time-at-sea data (paid providers: Spire, Kpler, MarineTraffic)
4. **Broker charter fixtures:** TC rates by route (proprietary market data)
5. **Free-float percentages:** FIF/IWF for accurate ATVR/FALR calculations
6. **Charterbook granular schedules:** Day-level remaining tenor for all vessels

20 Adapter Candidates

Adapter	Data Types	Must Have	Notes
sec_adapter	filings, 10-K/10-Q/6-K/8-K text	Yes	Charter mix, fleet, leverage, fuel disclosures
company_overview	company_profile, industry, topic_relevance	Yes	Universe cohorting, quick screening
polygon_adapter	prices, ADV, short_interest	Yes	Tradability, beta/risk, resilience overlay
fmp_adapter	earnings_transcripts, market_cap	No	Management commentary, fundamentals
patents_adapter	patent metadata	No	Not primary for this factor

Table 18: Adapter Candidates for Factor Implementation

21 Implementability Notes

21.1 Phase 1: Pre-Identified Sources

Pre-identified P&I circulars, MARAD advisories, and broker reporting are directly usable as web sources for trigger detection and mapping. SEC filings contain fleet and charter schedule disclosures but require LLM-assisted classification to extract %-spot exposure.

21.2 Phase 2: Autonomous Research Confirmation

Primary signals (P&I circulars, MARAD/IMO notices, broker premium reporting, AIS transit counts) are available publicly but often as unstructured prose or PDFs. SEC filings contain fleet and charter schedule disclosures requiring text classification.

21.3 Phase 3: Gap Analysis

Gaps (spot % and broker premium time-series, AIS) cannot be fully filled by existing adapters. Workarounds:

- Infer charter mix from filings and press releases
- Use short interest and price moves as proxies for market positioning
- Flag AIS and broker premium as external paid data to onboard

21.4 Phase 4: Adapter Mapping

All quantitative steps map to adapters: prices and short interest from `polygon_adapter`; filings from `sec_adapter`; company metadata from `company_overview`; earnings transcripts from `fmp_adapter`. Text-to-structured extraction implemented with `ClassificationAgent` and `LLM-ScoringAgent`.

21.5 Key Limitations

- No adapter for AIS/vessel-tracking or proprietary charter databases
- EWRI/reinsurance pricing terms are market/private data
- Web ingestion of club circulars required and must be kept updated
- Per-company charter splits must often be estimated from filings

22 Step-by-Step Implementation Guide

1. **Universe setup:** Load S&P 500 and Russell 2000; apply `company_overview` profiles and topic relevance to preselect shipping/marine-services and airlines/travel
2. **Tradability screen:** Compute 63-day ADV; require $ADV \geq \$5M$ and $price \geq \$5$; exclude OTC/Pink
3. **Filings ingestion:** Pull last 6–8 filings (10-K/20-F/6-K/10-Q/8-K) using `sec_adapter`
4. **Transcripts ingestion:** Fetch latest 4–8 earnings transcripts via `fmp_adapter`
5. **Shipping LLM scoring (A,B,D,E):** Score percent on spot, pass-through strength, fleet metrics, long-term TC share
6. **Balance-sheet resilience (C):** Extract leverage/liquidity signals; combine with short interest `days_to_cover`
7. **Airlines/travel scoring (F1,F2):** Extract fuel % of opex and hedge ratio; classify disruption sensitivity
8. **Assemble feature DataFrame:** Outer-join LLM scores and market features by ticker
9. **Compute composite Score:** Apply cohort-wise z-scores and dimension weights
10. **Risk model & betas:** Compute individual OLS betas vs SPY and shrinkage covariance
11. **Initial weights:** Compute base long/short leg weights; apply liquidity adjustment; clip to [0.5%, 7%]
12. **Constraint repair:** Enforce caps; renormalize; beta-neutralize
13. **Output & audit:** Generate constituent file, exposure report, backtest packs
14. **Rebalance & monitoring:** Re-run monthly; monitor ADV, betas, `days_to_cover`
15. **Post-trade checks:** Verify sector weights, zero-net beta, gross = 1.0

23 Citations

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13. Fama & French (2015) — A five-factor asset pricing model. <https://www.sciencedirect.com/science/article/abs/pii/S0304405X14002323>
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17. LNG vessels transits through Suez Canal under changing geopolitical context. <https://jshippingandtrade.springeropen.com/articles/10.1186/s41072-025-00205-3>
18. Impact of geopolitical risk on oil prices and freight rates. <https://www.sciencedirect.com/science/article/pii/S0360544223001731>

24 Document Metadata

Field	Value
Extraction Timestamp	2026-03-02T22:27:26 UTC
Pipeline Version	4.0.0
Documents Processed	228
Total Snippets Extracted	460
High-Confidence Threshold	0.50
Snippets After Filtering	279
Classification Model	gpt-5-mini
Agent Extraction Model	gpt-5-mini

Table 19: Extraction Metadata

A Short Leg Candidates: Airlines/Travel Fields

Fields to extract for short leg candidates:

- company_name, ticker, sector
- jet_fuel_cost_share_percent
- hedging_ratio_percent_next_12m
- long_haul_share_percent
- key_route_exposure_regions
- disruption_sensitivity_notes

- load_factor_percent_recent
- unit_revenue_rasm_casm_delta_bps
- guidance_revision_text_snippet
- net_debt_to_ebitda
- liquidity_headroom_usd
- operational_disruption_flags
- inclusion_rationale

B Fleet/Vessel Level Characteristics

Fields to extract for fleet analysis:

- owner_company, vessel_name, imo_number
- vessel_type, dwt_or_cbm, year_built
- fuel_type, scrubber_fitted, ice_class_flag
- max_service_speed_knots
- current_employment_status
- redelivery_window_date
- availability_within_30d

C Industry/Company Exposure Fields

Fields to extract for company-level analysis:

- company_name, ticker, report_date
- percent_fleet_on_spot
- percent_on_time_charter
- short_charter_ratio
- weighted_avg_tc_duration_days
- tce_usd_per_day_latest
- cash_and_equivalents_usd
- total_debt_usd
- available_credit_facilities_usd
- leverage_ratio
- fleet_avg_age_years
- war_risk_passthrough_mechanisms
- classification_flag